

ZHIYU WANG

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Agents LLMs AI4Bio Geometric DL



🏛️ EDUCATION

University of Cambridge

Advanced Computer Science, Master of Philosophy (Distinction, Rank: 8/60)

Cambridge, UK

Oct 2024 — Jul 2025

— Awarded Highly Commended MPhil Project Prize 2024–2025 by the Department of Computer Science and Technology

University College London

Computer Science, Bachelor of Science (First Class Honors)

London, UK

Sep 2021 — Jun 2024

🔬 RESEARCH EXPERIENCE

Auto-Research for the Virtual Cell Challenge | SAPIENT INTELLIGENCE & SJTU

Remote

Project lead | *Collaborator: Prof. Liang Hong*

Aug 2026 — Present

— Leading a joint project with Shanghai Jiao Tong University (SJTU) that applies Praxist, our AI-scientist framework, to zero-shot prediction of CRISPRi knockdown responses in the Virtual Cell Challenge 2026; ranked 167/1,169 (top 15%) on the preliminary leaderboard.

Bayesian Optimization for Auto-Research Systems | MILA – QUEBEC AI INSTITUTE

Remote

Collaborator | *Supervisor: Prof. Jian Tang*

Apr 2026 — Present

— Bayesian optimization for agentic evolution: a Gaussian process predicts expansion gains; Thompson sampling picks the next experiment.
— Best average rank on 15 AlphaEvolve tasks and top gold-medal rate (45%) on MLE-bench Lite.

Protein Substructure Alignment via Optimal Transport | UNIVERSITY OF CAMBRIDGE & SJTU

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò & Prof. Liang Hong*

Apr 2025 — Sep 2025

— Developed PLASMA, a plug-and-play optimal-transport module that adds fast, accurate local substructure alignment to any protein representation model with little or no training. ICLR 2026 poster. [\(Paper\)](#)
— Improved ROC-AUC by 10–30% across 7 protein language models on 3 datasets curated from InterPro (motifs, active/binding sites).
— Showed with PyMOL case studies that the aligned substructures are structurally similar, not artifacts of sequence similarity.

Hierarchical Protein Representation Learning | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò*

Oct 2024 — Jul 2025

— Built a complete SE(3)-equivariant framework that adds a secondary-structure level to protein representations via a hierarchical hypergraph, allowing a coarser-grained representation that gives better global understanding. [\(Invited Talk\)](#) [\(Paper\)](#)
— Leveraged this new structural hierarchy to achieve the best SCOP fold classification accuracy (43.3% vs. 38.4% for the same network without it; 40.1% best baseline), confirming the improved global understanding of protein structure.

Multi-Omics Data Integration with GNNs | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò*

Feb 2025 — Sep 2025

— Built MoRE-GNN, a heterogeneous graph autoencoder (GCN + GATv2) that learns relational graphs from single-cell RNA, protein and ATAC data; beats MOJITOO on BM-CITE (ARI 0.892 vs 0.868) and LUNG-CITE clustering and supports cross-modal prediction. [\(Paper\)](#)

Graph-based Pain Estimation from Facial Expressions | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Hatice Gunes*

Oct 2024 — Jan 2025

— Proposed GraphAU-Pain, which fuses a ResNet-50 backbone with graph-based facial action unit (AU) representations for interpretable pain intensity estimation; a transfer learning strategy for domain shift and missing labels improved AU-occurrence F1 by 30% on UNBC. [\(Paper\)](#)

Readmission Prediction with Remote Patient Monitoring Data | UNIVERSITY COLLEGE LONDON

London, UK

Project lead | *Supervisor: Prof. Ivana Drobnjak*

May 2023 — Jun 2024

— Built a SQL data pipeline and interpretable ML models (ROC-AUC 0.79±0.03, SHAP) on remote patient monitoring data with UCL Hospital clinicians, using patient-independent stratified bootstrapping to handle extreme data scarcity (55 patients).

INDUSTRIAL EXPERIENCE

Autonomous Research Agents | SAPIENT INTELLIGENCE

ML Research Scientist

Beijing, China

Sep 2025 — Present

- Developing **Praxist**, an autonomous AI-scientist framework with DAG-based execution lineage for solution provenance; achieved ~30% higher performance at 10% of the cost of Claude Code with Opus on Karpathy's AutoResearch prompt.
- Showcased Praxist at **AI4 2026** through live booth demos and conversations with prospective customers and collaborators.

Healthcare LLM Knowledge Distillation | XUNFEI HEALTHCARE TECHNOLOGY

Intern ML Researcher

Hefei, China

Jul 2024 — Sep 2024

- Distilled a teacher LLM into synthetic medical instruction data, raising **Xiaoyi**'s instruction-following accuracy from 34% to 82% via SFT.

RAG-Based Financial Intelligence System | LUOJIN DATA INFORMATION

Intern ML Engineer

Shanghai, China

May 2023 — Aug 2023

- Built a RAG pipeline (MongoDB, LangChain) for financial report classification with 97% entity matching.

Homepage Navigation Agent for Hospitals | UCL & NATIONAL HEALTH SERVICE

Industrial Collaboration Program

London, UK

Sep 2022 — May 2023

- Built a system that crawls a hospital's website and auto-generates an agent that answers visitor questions and guides them to the right pages, cutting setup from weeks of manual development to 30 minutes; demoed at Great Ormond Street Hospital ([media coverage](#)).

PUBLICATIONS

Conference, Journal & Workshop Papers

* equal contribution

1. "Fast and Interpretable Protein Substructure Alignment via Optimal Transport."

Zhiyu Wang*, Bingxin Zhou*, Jing Wang, Yang Tan, Weishu Zhao, Pietro Liò, Liang Hong. *ICLR*, 2026. ([Paper](#))

2. "Data-Efficient AU-Graph Transfer Learning for Consumer-Camera Pain-Aware Sensing."

Zhiyu Wang, Han Wei Wang, Hangqian Li, Jun Jiang, Hatice Gunes, Yang Liu. *IEEE Consumer Electronics Magazine*, 2026. ([Paper](#))

3. "GraphAU-Pain: Graph-based Action Unit Representation for Pain Intensity Estimation."

Zhiyu Wang, Yang Liu, Hatice Gunes. *IJCAI 2025 MiGA Workshop*. ([Paper](#))

Preprints

4. "Praxist: From Experimental Artifacts to Solution Lineages."

Jin Li*, Ahmed Murtadha*, **Zhiyu Wang***, Qiwen Chen*, William Chen*, Yifei Wu*, Guan Wang*, et al. *arXiv*, 2026. ([Paper](#))

5. "Topotein: Topological Deep Learning for Protein Representation Learning."

Zhiyu Wang, Arian Jamasb, Mustafa Hajj, Alex Morehead, Luke Braithwaite, Pietro Liò. *arXiv preprint*, 2025. ([Paper](#))

6. "MoRE-GNN: Multi-omics Data Integration with a Heterogeneous Graph Autoencoder."

Zhiyu Wang, Sonia Koszut, Pietro Liò, Francesco Ceccarelli. *arXiv preprint*, 2025. ([Paper](#))

ADDITIONAL EXPERIENCE

- **Reviewer Experience:** NeurIPS 2026; AI4Science Workshop at NeurIPS 2025 and ICML 2026
- **Teaching Experience:** Programming Tutor at UCL (2022–23)